

ICT702 Data Visualisation

Visualising Variability

Module 5



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



Part 3

Measures of Central Location: Definitions

Definitions: **Mean**

- Sum of the values divided by the sample size
- AVERAGE function in Excel

Median

- Sort the data in ascending order:
 - If the sample size is even, take the average of the two middle points
 - If the sample size is odd, take the middle value
- MEDIAN function in Excel

Mode

- Most frequent value(s) in the data set
- MODE.MULT function in Excel

Measures of Central Location: Application

	A	B	C	D
1	Selling Price			
2	108000		Mean: =AVERAGE(A2:A13)	
3	138000		Median: =MEDIAN(A2:A13)	
4	138000		Mode 1: =MODE.MULT(A2:A13)	
5	142000		Mode 2:	
6	186000			
7	199500			
8	208000			
9	254000			
10	254000			
11	257500			
12	298000			
13	456400			

	A	B	C	D
1	Selling Price			
2	\$108,000		Mean:	\$219,950
3	\$138,000		Median:	\$203,750
4	\$138,000		Mode 1:	\$138,000
5	\$142,000		Mode 2:	\$254,000
6	\$186,000			
7	\$199,500			
8	\$208,000			
9	\$254,000			
10	\$254,000			
11	\$257,500			
12	\$298,000			
13	\$456,400			

Measures of Variability: Range and Standard Deviation

Definitions:

Range

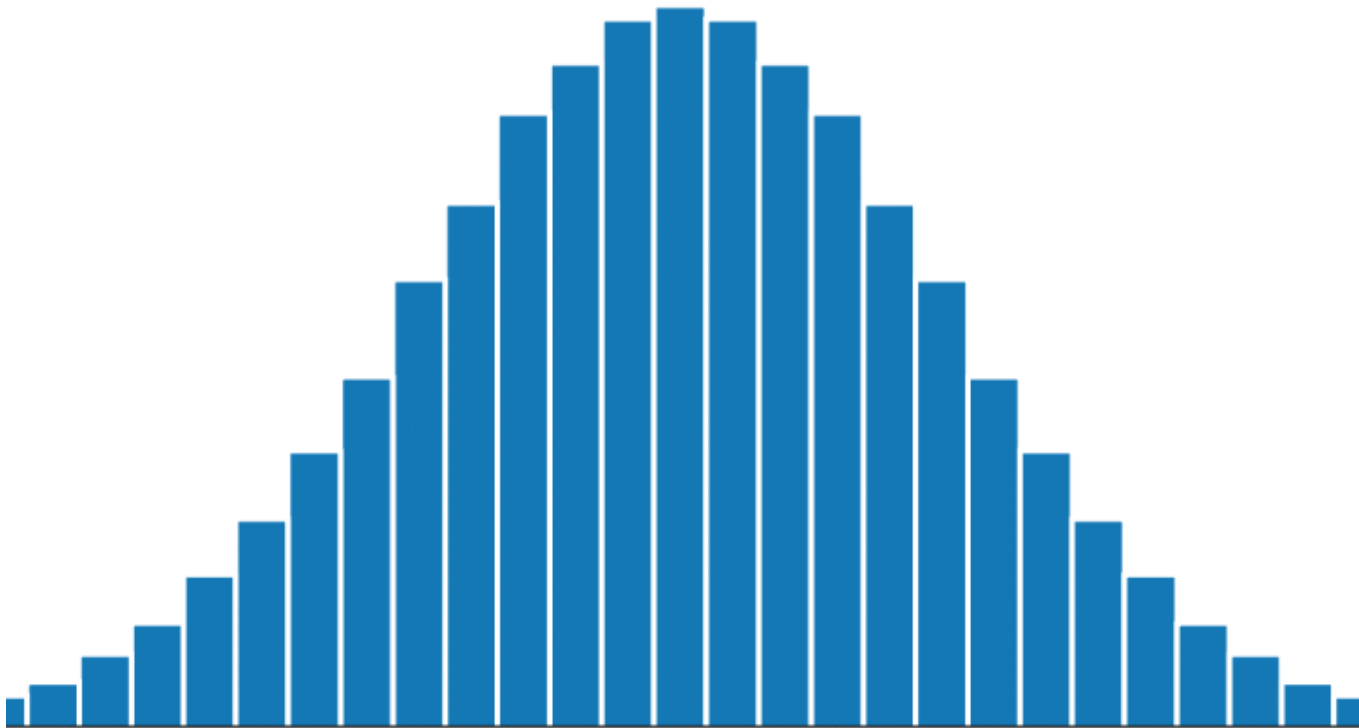
- Largest value minus smallest value in the set
- MAX minus MIN functions in Excel

Standard Deviation

- Based on average deviation from the mean
- STDEV.S function in Excel

The Empirical Rule for Bell-Shaped Distributions

Bell-shaped distribution (symmetric, single-mode)



Using the standard deviation to describe variability:

≈ 68% of data values lie within one standard deviation of the mean

≈ 95% of data values lie within two standard deviations of the mean

≈ 99.7% of data values lie within three standard deviations of the mean

Measures of Variability: Percentiles

Definitions: **Percentile**

- The p^{th} percentile is a value that exceeds $p\%$ of the observations in the set
- PERCENTILE.EXT function in Excel

Quartile

- $Q1 = 1^{st}$ quartile = 25^{th} percentile
- 2^{nd} quartile = 50^{th} percentile = median
- $Q3 = 3^{rd}$ quartile = 75^{th} percentile

Interquartile Range (IQR)

- $Q3 - Q1$
- 75^{th} percentile minus 25^{th} percentile

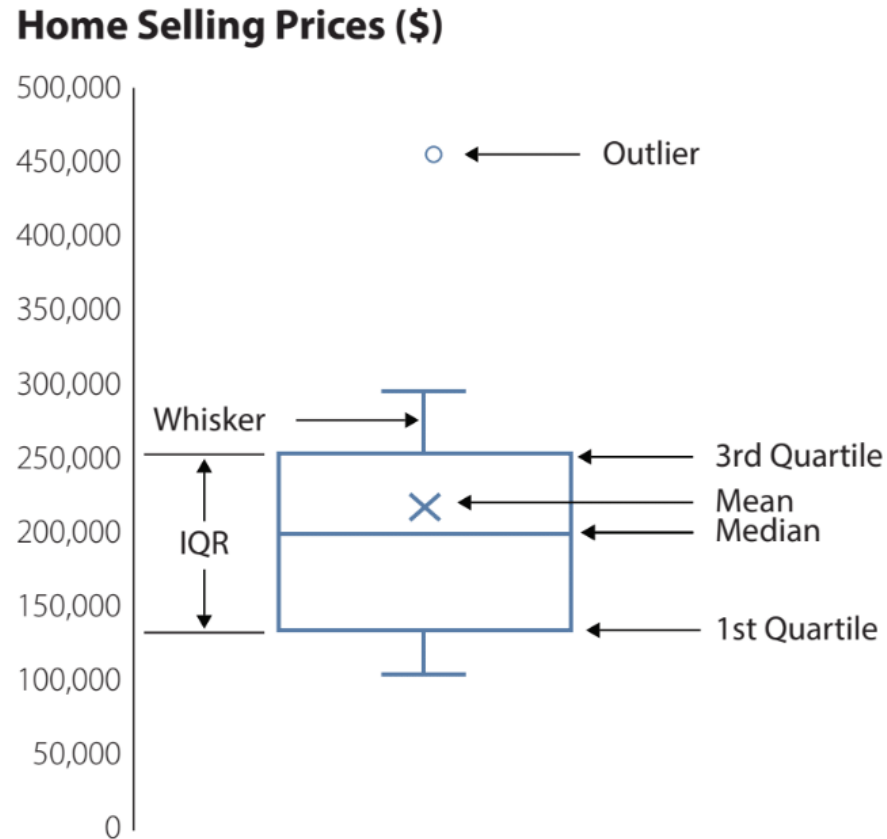
Measures of Variability: Application

	A	B	C	D
1	Selling Price			
2	108000		Mean: =AVERAGE(A2:A13)	
3	138000		Median: =MEDIAN(A2:A13)	
4	138000		Mode 1: =MODE.MULT(A2:A13)	
5	142000		Mode 2:	
6	186000			
7	199500		Range: =MAX(A2:A13) - MIN(A2:A13)	
8	208000		Standard Deviation: =STDEV.S(A2:A13)	
9	254000			
10	254000		25th Percentile: =PERCENTILE.EXC(A2:A13,0.25)	
11	257500		50th Percentile: =PERCENTILE.EXC(A2:A13,0.5)	
12	298000		75th Percentile: =PERCENTILE.EXC(A2:A13,0.75)	
13	456400			
14			IQR: =D12-D10	

	A	B	C	D
1	Selling Price			
2	\$108,000		Mean: \$219,950	
3	\$138,000		Median: \$203,750	
4	\$138,000		Mode 1: \$138,000	
5	\$142,000		Mode 2: \$254,000	
6	\$186,000			
7	\$199,500		Range: \$348,400	
8	\$208,000		Standard Deviation: \$95,100	
9	\$254,000			
10	\$254,000		25th Percentile: \$139,000	
11	\$257,500		50th Percentile: \$203,750	
12	\$298,000		75th Percentile: \$256,625	
13	\$456,400			
14			IQR: \$117,625	

Box and Whisker Charts: Definitions

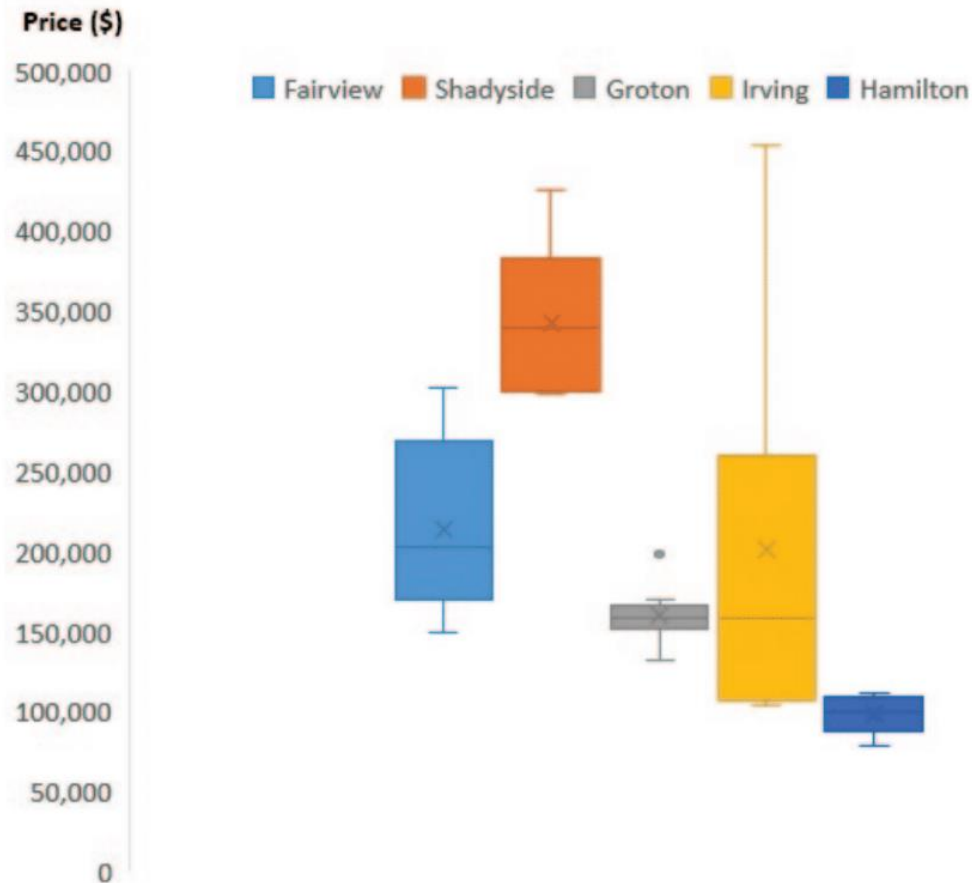
Box and whisker chart for *CincySales* data



Outliers are values outside the range:
 $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$

Box and Whisker Chart to Compare Multiple Variables

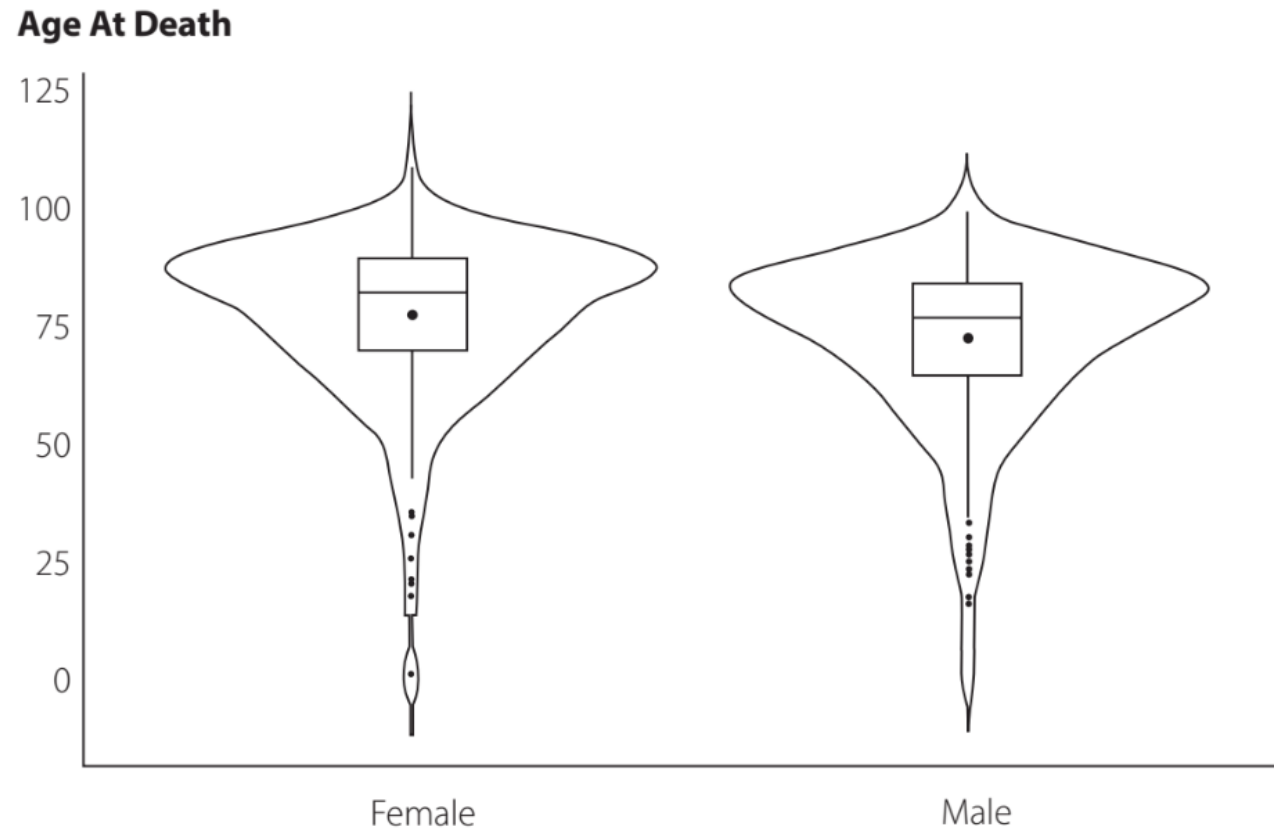
Comparing Home Selling Prices in Different Suburbs



Observations:

- Shadyside has the highest home selling prices, while Hamilton has the lowest.
- Irving and Groton have the same median selling price, but Irving houses have a higher price variability.

Additional charts: the Violin Chart



A **Violin chart** is an advanced visualisation that combines the statistical descriptors of a box and whisker chart with a rotated and mirrored kernel density chart.

End of Part 3

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